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Legality of LLMs in the Autonomous Driving Space

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I. Abstract

Given that the automotive and autonomous automotive sectors are highly regulated, a vehicle must always comply with all current regulations or face the risk of non-compliance. With laws and regulations continually evolving at both the state and national levels, ensuring that a vehicle complies with all applicable requirements at all times is challenging. However, this difficult task can be significantly easier with an LLM tailored to interpret all regulations and generate developer-understandable recommendations, which can then be implemented in the ADAS system. Overall, this LLM is expected to significantly shorten the developer timeline and reduce the risk of non-compliance, with the only major risk being that the LLM user is responsible for all implemented outputs.

II. Background Introduction

2.1 Introduction to SAE Driving Levels

SAE International defines six levels of driving automation in SAE J3016. These levels are characterized by how responsibilities are divided between the human driver and the automated system. The framework uses key elements such as:

- Dynamic Driving Task (DDT) – real-time operational control of the vehicle
- Object and Event Detection and Response (OEDR) – detecting and responding to hazards
- DDT fallback – response when the system reaches its limits
- Operational Design Domain (ODD) – specific conditions under which automation is intended to function

The levels range from Level 0 (no automation) to Level 5 (full automation):

Level	Description
L0	The driver performs the entire DDT. The system may provide warnings or momentary interventions (e.g., emergency braking), but no sustained control.
L1	The system continuously performs either lateral control (steering) or longitudinal control (acceleration/braking) within a defined ODD. The driver performs all remaining aspects of DDT and OEDR.
L2	The system continuously performs both lateral and longitudinal control within a defined ODD. The driver remains responsible for OEDR, monitoring the system, and performing DDT fallback when required.
L3	The automated driving system performs the entire DDT within a defined ODD. A takeover-ready user must respond to system requests to intervene.
L4	The system performs the entire DDT and fallback within a defined ODD. No user intervention is expected during normal operation.

L5 The system performs the entire DDT and fallback in all environments and conditions, without ODD limitation. No user intervention is expected.

2.2 Definition and functions of ADAS for L2

From an SAE classification perspective, the defining characteristic of Level 2 (Partial Driving Automation) is not the number of features provided, but the allocation of responsibility between the system and the human driver. At this level, the system is responsible only for the sustained execution of lateral and longitudinal motion control within a defined Operational Design Domain (ODD). The driver, however, remains responsible for Object and Event Detection and Response (OEDR), must continuously supervise the driving environment and system performance, and must immediately intervene when required. In other words, the Dynamic Driving Task (DDT) fallback remains the driver's responsibility at all times. The U.S. National Highway Traffic Safety Administration (NHTSA) characterizes Level 2 as "You drive, you monitor," emphasizing that although the system can simultaneously assist with steering and acceleration or deceleration, the driver must remain actively engaged and attentive to the roadway environment.

In practical implementation, Level 2 systems typically operate through a closed-loop control architecture centered on maintaining lane position (lateral control) and regulating vehicle speed and following distance (longitudinal control). Commercially deployed systems often rely on radar, camera-based machine vision, and sensor fusion technologies to enhance robustness and stability. However, increased sensor capability or algorithmic sophistication does not alter the fundamental division of labor: the driver retains responsibility for environmental monitoring and hazard response.

Level 2 systems commonly integrate features such as Adaptive Cruise Control (ACC), which maintains speed and following distance, and Lane Centering or Lane Keeping Assist, which maintains lane position. When these two functions operate simultaneously and continuously, the system meets the criteria for Level 2 classification; when only one operates continuously, it is typically categorized as Level 1. Many vehicles equipped with Level 2 automation also include Lane Change Assist that performs a single lateral maneuver upon driver initiation or confirmation, as well as intervention-based safety functions such as Blind Spot Intervention and Automatic Emergency Braking (AEB), which can apply steering or braking to mitigate imminent risks. While these features enhance convenience and provide additional safety layers, they do not replace the driver's obligation to monitor the environment and maintain ultimate control of the vehicle.

Within Level 2 systems, conversational interaction should be understood as structured, task-focused human–machine interface (HMI) communication rather than human-like dialogue. Such interactions are designed to support accurate driver mental models, reinforce system boundaries, maintain situational awareness, and reduce misuse or over-reliance. Voice commands are typically limited to short, low-risk instructions with explicit confirmation feedback, while physical controls enable mode switching and parameter adjustments and are clearly separated from safety-critical functions. Multimodal alerts—including visual, auditory, and haptic signals—communicate system status, limitations, and disengagements in a manner that is salient for drivers who are required to remain in a supervisory role. System perception and computational reasoning are narrowly scoped to support tactical vehicle control,

such as lane tracking and speed regulation, rather than comprehensive scene interpretation. Reliability is therefore contingent upon operation within the defined ODD and on active driver supervision. Performance may degrade near ODD boundaries, under complex environmental conditions, or during periods of high interaction load, where failures can occur with limited warning and require immediate driver intervention.

III. Concepts and Uses of LLM and NLP

3.1 Definition and applications of NLP

Natural Language Processing (NLP) researches technologies that enable computers to represent, understand, and generate human language. Core tasks include word segmentation, part-of-speech tagging, named entity recognition, dependency parsing, semantic understanding, and dialogue systems; typical applications include search/ranking, customer service and information extraction, machine translation, and voice assistants. In driving scenarios, NLP is used for understanding in-vehicle voice commands (intent/slot), clarifying and correcting interactions, and monitoring driver-related language/dialogue interactions and prompts.

3.2 Definition and applications of LLM

Large Language Models (LLMs) typically refer to general-purpose language models that are pre-trained on massive corpora and possess strong contextual modeling and generation capabilities. After fine-tuning/alignment with instructions, they can act as dialogue agents to perform tasks such as text generation and rewriting, instruction following, code generation, and knowledge retrieval enhancement. In automotive/ADAS systems, LLMs can improve the multi-turn understanding of voice assistants, explain system status/limitations and operating procedures, and conduct Q&A in the owner's manual. However, they are constrained by real-time performance and computing power, interpretability and consistency, data privacy and compliance, and security (illusion and unauthorized execution), requiring strict permissions, verification, and human factors design.

IV. Laws surrounding LLMs and SAE L2

4.1 Current laws surrounding LLMs and NLPs

(a.) Regulatory Landscape (Overall)

- Legal frameworks for LLMs and NLP systems are still evolving
- Regulations focus on accountability, transparency, and risk governance
- No federal law yet

(b.) Current liability allocation for LLM systems :

- User Responsibility
    - Users are responsible if they misuse or over-reli on LLM outputs
    - Liability may apply when:
        - Outputs are implemented without verification
        - Disclaimers or usage limitations are ignored
        - AI is used beyond its intended scope

- AI Model Provider Responsibility
    - Model developers may share liability if:
        - Systems produce systematically harmful outputs
        - Safety guardrails are insufficient
        - Known risks were not disclosed
- Deploying Company Responsibility
    - Companies integrating LLMs (e.g., ADAS developers) typically bear primary liability because they:
        - Control deployment context
        - Validate outputs
        - Make final engineering decisions

4.2 Current laws surrounding L2 Driving

(a.) Regulatory Landscape (Overall)

- Level 2 (L2) systems are legally classified as driver assistance, not autonomous driving
- The human driver remains fully responsible for vehicle operation at all times
- Regulations focus on:
    - Driver supervision
    - System operational boundaries
    - Safe human–machine interaction
- Drivers must remain attentive and ready to intervene immediately when ADAS is active.

(b.) Current related laws and regulations: In the U.S., liability frameworks for LLMs are not yet codified in a single law but are instead derived from existing regulatory enforcement and emerging AI governance frameworks.

- Vehicle safety is regulated federally by
  National Highway Traffic Safety Administration (NHTSA)
- L2 systems must comply with existing Federal Motor Vehicle Safety Standards (FMVSS)
- Focus areas include:
    - Braking performance
    - Steering control
    - Crash avoidance technologies
    - Human–machine interface safety

V. Very Basic Initial Model: Using an LLM Prototype in the Development of ADAS

5.1 Conceptual architecture description

The goal of the created LLM is to ingest state laws about L2 autonomous vehicles and output a summary. This summary should be clear enough for a software engineer to interpret into behavioral

requirements for the Advanced Driving Assistance System. For example, L2 is defined as having the car control, steering, and speed. However, the human driver must be able to take over instantly if something goes wrong. The output of this would be:

"To abide by the SAE levels of driving automation, it is highly advised to have the car prompt the driver whenever their hands are not detected on the steering wheel and when their eyes are off the road for more than five seconds. This would ensure the vehicle follows the L2 SAE regulations that 'the driver must constantly monitor the system'."

The LLM will output clear suggestions and requirements in plain English, not finished code. While this LLM's goal is to simplify the understanding of laws and their interaction with ADAS, developers must still understand the main point of the laws. This ensures that their vehicles are abiding by them. If the LLM only outputs code but does not summarize how the code helps the company follow the law, there is a high risk of non-compliance. LLMs can make mistakes and misinterpret ambitious or complex legal documents. Having a developer interpret the LLM's output and suggestions reduces the risk of misinterpretation. This ensures a human is constantly checking that the AI's output matches the inputted laws.

It is important to note that due to the complexity and ambiguity of laws and regulations, this LLM needs significant supervision by humans for at least the first few months of implementation. A person will have to ensure that the LLM correctly interprets the laws that it was fed and that its suggested behavioral outputs are in scope of ADAS, and more importantly, will ensure that the recommendation will actually ensure that the intended law is followed. Once the LLM is consistently correct in its interpretation of L2 laws, and its recommendations do improve the ADAS, then human supervision can be relaxed.

Because a company that uses an LLM is held responsible for any incorrect behavioral requirements suggested by the LLM, should a vehicle be found non-compliant, the users must understand they are responsible for anything they choose to implement based on the LLM's output. This will be addressed when moving from the initial model to the final product. Once the final model is ready for launch, it will require users to thoroughly read and understand a set of terms and conditions that outline how they are at fault if anything goes wrong. Every user must sign this agreement every time they start a new session with the LLM to signify that they understand the risk they are taking using this LLM as part of the development process. Additionally, two-factor authentication will be required upon the start of every session with the LLM to ensure only the appropriate user is using this tool. To uncover the optimal rate at which the user has to use two-factor authentication and re-sign the terms and conditions agreement, the company would benefit from using UI experimentation. Using experimentation allows the company to strike the right balance between having barriers to sign-in and ensuring the user thoroughly understands the risk of using this AI.

5.2 Net positives from LLM Prototype to help with understanding laws

With this LLM, the timeline from when laws are published to updating ADAS will shorten. Lawyers already see how AI reduces contract review time. Agreements that took days now take hours (2026 Priore). This lets autonomous car companies cut the time between receiving new ADAS laws and getting products on the road. For example, if it traditionally takes 3 days to review a law and integrate it into the ADAS, with this LLM model, this can be reduced to just 4 hours, 18 times faster than before. This compression of the development timeline means laws are met at all times at a significantly faster

rate, and developers can focus on other, more innovative ADAS projects instead of compliance issues.

Additionally, using an LLM to understand laws bridges the gap between developers and lawyers. Traditionally, it is assumed that only lawyers need to understand the laws, and developers just write the code. Ideally, both lawyers and developers understand the law to ensure that both groups converge on an idea that will ensure ADAS always follows the law, and the improvement is feasible for implementation into ADAS. This is possible with an LLM that ingests state laws and outputs a summary that can be interpreted into behavioral requirements, as developers can now understand the key points of the law without having to scrub through pages of legal jargon. By using this LLM, both lawyers and developers can come together to understand what the laws are asking of them to get to actionable requirements that developers can then implement. With both parties more informed, this minimizes the risk of non-compliance and misunderstanding.

Since driving across state lines is extremely common, it is important to ensure that ADAS follows all laws in all states, not just the state where the vehicle was developed. Using an LLM like the one proposed allows easy state-by-state comparisons to ensure that ADAS abides by even the most strict laws of each state. Having a legal team keep track of, interpret, and summarize all laws in every state to ensure that software developers correctly design ADAS is impossible, as rules and regulations are constantly evolving. However, LLMs eliminate this issue as they can ingest all state laws in seconds and output actionable information that ensures ADAS follows even the strictest of laws. This way, drivers can drive wherever they please without worrying about non-compliance issues.

5.3 Risks of using an LLM Prototype to understand laws

While the timeline reduction going from new laws surrounding autonomous driving to integrating behavioral requirements into the ADAS will improve greatly from this LLM, it is important to note that this increase in efficiency can only occur if the legal team first knows the types of laws that have the most risk associated with non-compliance (2026 Priore). Without people, what processes can be automated, and what processes have to be looked at by a person, using this LLM may slow productivity rather than increase it. Ideally, a company would need to hire a highly experienced lawyer to supervise how the LLM analyzes different types of laws and provide feedback on how well its suggestions comply with current laws. This would initially cost a company a significant amount of both time and money, having this LLM is worth it in the long run, as it has the potential to greatly decrease the amount of time it takes to ensure all vehicles everywhere are compliant with the law.

Unfortunately, LLMs have the potential to “hallucinate” facts and primary sources. This poses a major risk in terms of law compliance for the ADAS and begs the question, “Whose fault is it when AI makes a mistake?” In 2023, a personal injury lawyer relied heavily on ChatGPT to help him prepare a court filing and failed to check over cases, rulings, and direct quotes, which were found to be completely made up by the LLM (2023, The Economist). In this case, the lawyer was found to be at fault for not checking over the AI’s work, not the AI’s fault for producing incorrect information (2023, The Economist). In this sense, a company that implements incorrect behavioral requirements into its ADAS would be found at fault if the vehicle was found non-compliant with the law, not the AI that suggested the behavioral requirements. This puts significant pressure on any company that uses the proposed LLM to ensure that they are rigorously checking over the AI’s suggestions before implementing them.

Ideally, this LLM would function as a live, interactive Wikipedia page rather than a conversation

friend to mitigate the risk of hallucinations. LLMs are designed to be excessively helpful and agreeable rather than producing accurate results; thus, causing the model to output inaccurate results, known as hallucinations (2025, Chen et al). Because the autonomous vehicle space is heavily regulated for the safety of all people, minimizing the risk of hallucinations given by the LLM is extremely important. Having the LLM function as a Wikipedia page rather than a conversation friend allows it to prioritize accuracy of information rather than helpfulness, which leads to fewer overall hallucinations. For example, when medical researchers blatantly prompted multiple LLMs to reject any illogical prompts given rather than produce inaccurate results, in 94% of cases, GPT models rejected illogical requests and correctly identified the reason for the rejection instead of outputting misinformation (2025, Chen et al). When researchers did not give the LLMs explicit permission to reject an illogical request, LLMs would produce misinformation 100% of the time (2025, Chen et al). This stark difference in behavior is key to ensuring that the LLM is as accurate as possible with its information to always ensure compliance with the law.

VI. Market Analysis incorporating a Legal LLM

6.1 Target customers and value proposition

A legal LLM for ADAS compliance primarily targets organizations that must repeatedly translate evolving laws, guidance, and standards into traceable engineering requirements and audit-ready documentation. Core customer segments include: (1) OEM ADAS and compliance teams seeking faster conversion of regulatory updates into implementable requirements; (2) Tier-1 suppliers and ADAS software vendors seeking reusable “compliance language” across multiple OEMs and platforms; (3) corporate legal/compliance departments seeking to reduce time spent reviewing long documents while retaining human approval; and (4) fleets and mobility operators seeking clearer, region-specific policies and training materials to reduce misuse and standardize operations. This aligns with broader legal-industry adoption patterns where individual experimentation with generative AI is increasing, but organization-wide adoption remains constrained by governance and policy requirements.

6.2 Customer preferences and acceptance

Across these segments, adoption is most plausible when the system is positioned as a compliance assistant (summarize, compare, draft, and organize evidence) rather than a legal decision-maker. Buyers typically prefer enterprise controls that support verification and accountability, including traceable citations and workflow integration, because legal and safety contexts are highly sensitive to errors and reputational risk. Survey evidence suggests firms prioritize tools that integrate into legal workflows and ethics constraints, and that adoption can be slowed by restrictive policies and ethical concerns.

Concerns about hallucinations are not theoretical: U.S. courts have sanctioned lawyers for submitting filings containing AI-fabricated or misrepresented citations and facts, reinforcing the market demand for verification, auditability, and human oversight.

In addition, vendor usage should be considered explicitly. Many firms in regulated industries prefer not to build a foundation model from scratch, and instead adopt an existing vendor model or an

open-weight base model that can be integrated into their own workflow. For Latitude AI, a practical option would be to use a free or open Hugging Face base model as the starting point, then add company-specific retrieval, citations, logging, and review controls on top of it. Hugging Face documentation shows that models can be downloaded and integrated into existing libraries, which makes this a realistic starting point for an internal compliance assistant rather than a full model-development effort from zero. This approach can improve control and reduce dependence on a single commercial vendor, while still preserving the enterprise safeguards buyers expect in legal and safety-sensitive settings.

6.3 Cost of producing the product and build-vs-buy decision

The cost of a Legal LLM for ADAS compliance is best understood as the total cost of ownership (TCO), not just model tokens. In practice, recurring cost comes from four components: (1) model inference, (2) retrieval and storage for regulatory sources, (3) enterprise governance controls (citations, logs, access controls, review workflows), and (4) ongoing human verification and evaluation. Governance is not optional in legal use: the EU AI Act's record-keeping requirements for high-risk systems explicitly emphasize lifecycle logging, reinforcing the market expectation for auditability. NIST similarly frames AI risk management as a lifecycle activity that requires continuous monitoring and accountability mechanisms.

Model inference costs scale with usage volume and chosen model capability. Public pricing illustrates large spreads: OpenAI lists GPT-5.2 at $1.75 / 1M input tokens and $14 / 1M output tokens, while GPT-5 mini is $0.25 / 1M input and $2 / 1M output. Anthropic reports Claude Opus 4.6 pricing at $5 / 1M input tokens and $25 / 1M output tokens. These figures show why annual spend can range from modest to substantial, depending on (1) how many documents are processed, (2) how long the prompts or outputs are, and (3) whether premium models are used for every task.

However, for a compliance product, non-model costs often dominate. Retrieval-based designs require vector storage and query tooling; for example, OpenAI prices File Search vector storage at $0.10 per GB per day and File Search tool calls at $2.50 per 1,000 calls. Cloud hosting can also introduce tier premiums: Amazon Bedrock notes that Priority tier pricing is a 75% premium over Standard for certain offerings, and it lists per-token rates that vary by provider, model, and region.

This motivates the key strategic choice: use another company's existing LLM vs. build in-house. A practical research conclusion is that buy is typically appropriate for an initial product (faster validation, best baseline capability), but it creates vendor risk: providers can change prices, modify privacy/retention restrictions, or alter terms and availability. The mitigation is architectural rather than purely contractual: build a model-agnostic governance layer (citations, logs, review approvals, evaluation) so the underlying model can be swapped across providers without rewriting the compliance workflow. "Build in-house" is rarely economical if it means training a foundation model from scratch; the more realistic in-house approach is hosting and controlling the compliance stack while treating the base model as replaceable.

For Latitude AI specifically, "build in-house" should not mean training a new large language model from the ground up. A more realistic in-house path would be to start from an existing open-weight model, such as one accessed through Hugging Face, and then build the compliance workflow around it. In

this case, the proprietary value would come less from the base model itself and more from the surrounding system: curated legal sources, retrieval pipelines, audit logs, traceable citations, review checkpoints, and integration into the ADAS development process. This makes the product more economically realistic while still allowing the company to reduce vendor lock-in.

Finally, is the expenditure worth it? The economic case should be framed as a measurable tradeoff between recurring costs (model, storage, engineering, review) and avoidable compliance burden (time spent reading new rules, drafting requirements, creating evidence packages, and reconciling cross-state differences). The demand for verification is reinforced by real legal incidents: courts have sanctioned lawyers for filings containing AI-generated hallucinated or fabricated citations, which strengthens buyer insistence on human review, traceability, and accountability—features that are costly but necessary. Therefore, the product is "worth it" only if pilots demonstrate consistent reductions in (1) time-to-update after rule changes, (2) rework caused by unclear requirements, and (3) audit preparation effort—while maintaining robust verification before anything becomes an engineering requirement.

VII. Appendix

7.1 Sources

Amazon Web Services. "Amazon Bedrock Pricing" AWS,

n.d.,https://aws.amazon.com/bedrock/pricing/.DOI: N/A. Accessed 23 February 2026.

Anthropic. "Claude Opus 4.6" Anthropic News,

n.d.,https://www.anthropic.com/news/claude-opus-4-6.DOI: N/A. Accessed 23 February 2026.

Chen S *et al.* "When Helpfulness Backfires: LLMs and the Risk of False Medical Information Due to

Sycophantic Behavior" *npj Digital Medicine* 17 October 2025,

https://www.massgeneralbrigham.org/en/about/newsroom/press-releases/large-language-models-prioritize-helpfulness-over-accuracy-in-medical-contexts#:~:text=A%20new%20study%20led%20by,information%20necessary%20to%20do%20so.

DOI: 10.1038/s41746-025-02008-z. Accessed 22 February 2026

European Commission (AI Act Service Desk). "Article 12: Record-keeping" EU Artificial Intelligence

Act Service Desk, n.d., https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-12. DOI: N/A.

Accessed 23 February 2026.

Merken S. "New York lawyers sanctioned for using fake ChatGPT cases in legal brief" Reuters, 22 June

2023, https://www.reuters.com/legal/new-york-lawyers-sanctioned-using-fake-chatgpt-cases-legal-brief-2023-06-22/.DOI: N/A. Accessed 23 February 2026.

OpenAI. “API Pricing” OpenAI, n.d., https://openai.com/api/pricing/.DOI: N/A. Accessed 23 February 2026.

Priore, Ken. “The timeline compression that hit software development is here for legal work.” *LinkedIn*, 21 January 2026, https://www.linkedin.com/pulse/timeline-compression-hit-software-development-here-legal-ken-priore-msoie/. Accessed 15 February 2026.

Raymond N. “US appeals court orders lawyer to pay $2,500 over AI hallucinations in brief” Reuters, 18 February 2026, https://www.reuters.com/legal/government/us-appeals-court-orders-lawyer-pay-2500-over-ai-hallucinations-brief-2026-02-18/.DOI: N/A. Accessed 23 February 2026.

Tabassi E. “Artificial Intelligence Risk Management Framework (AI RMF 1.0)” National Institute of Standards and Technology, January 2023, https://doi.org/10.6028/NIST.AI.100-1.DOI: 10.6028/NIST.AI.100-1. Accessed 23 February 2026.

The Economist. “Generative AI could radically alter the practice of law.” *The Economist*, 6 June 2023, https://www.economist.com/business/2023/06/06/generative-ai-could-radically-alter-the-practice-of-law. Accessed 15 February 2026.

United States Court of Appeals for the Fifth Circuit. “No. 25-20086 (Order addressing AI hallucinations in a brief and sanctions)” 18 February 2026, https://www.ca5.uscourts.gov/opinions/pub/25/25-20086-CV0.pdf. DOI: N/A. Accessed 23 February 2026.